

IHTC Problem Breakdown

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METAMODEL

Root / Container

Class: HospitalInstance

- Patients:[] ref- Patient
- Nurses:[] ref- Nurse
- Surgeons:[] ref- Surgeon
- Rooms:[] ref- Room
- Operating theatres:[] ref- OperatingTheatre
- decisionHorizon: int

Core Entities

Class: Patient

- id: int
- isMandatory: boolean (current workaround for matching based on Status)
- isScheduled: boolean (current workaround for comparison inside the predicate)
- dueDate: int
- releaseDate: int
- ageGroup: AgeGroup
- gender: Gender
- surgeryDuration: int
- stayLength: int
- incompatibleRooms:[] ref- Room
- assignedSurgeonId: ref- Surgeon

Class: Room

- id: int
- maxCapacity: int

Class: Nurse

- id: int
- skillLevel: int

Class: Surgeon

- id: int

Class: OperatingTheatre

- id: int

Temporal / Supporting Entities

Class: HospitalisationShift

- patientId: ref- Patient
- shift: int

- skillLevelRequired: int
- workload: int

Class: NurseRosterEntry

- nurseId: ref- Nurse
- shiftId: ref- HospitalisationShift
- maxWorkload: int

Class: SurgeonAvailability

- surgeonId: ref- Surgeon
- day: int
- maxOperatingTime: int

Class: OTAvailability

- OTid: ref- OperatingTheatre
- day: int
- maxCapacity: int

Class: RoomAvailability

- roomId: ref- Room
- day: int
- occupiedBeds: int

Class: DeletedAdmissionsTracker

- count: int

Decision / Assignment Entities

Class: Admission

- patientId: ref- Patient
- admissionDay: int
- roomId: ref- Room
- OTid: ref- OperatingTheatre
- isPostponed: ENUM YES/NO

Class: RoomShiftAssignment

- nurseId: ref- Nurse
- roomId: ref- Room
- shift: ref- HospitalisationShift

Boundary Data

Class: Occupant (subclass of Patient)

- patientId: ref- Patient
- assignedRoomId: ref- Room
- stayLength: int

PROBLEM

Decisions

- D1:** What day in the planning horizon will each patient be admitted?
- D2:** What operating theatre will each patient use?
- D3:** What recovery room will each patient use?
- D4:** What rooms will each nurse cover on their assigned shifts?

Model Transformations

0.1 Patient Admission and Removal

- Transformation 1: Insert a random non-mandatory patient with a random admission day, room, and theatre. Classes used: `Patient`, `Admission`, `Room`, `OperatingTheatre`. Purpose: to increase the number of scheduled patients and explore new possible allocations quickly. It is a broad exploratory move.
- Transformation 2: Insert the non-mandatory patient with the least workload into a feasible admission day, room, and theatre. Classes used: `Patient`, `Admission`, `Room`, `OperatingTheatre`, `SurgeonAvailability`, `OperatingTheatreAvailability`, `RoomAvailability`. Purpose: to add a patient who is easier to fit into the schedule.
- Transformation 3: Find the earliest available surgeon, then find a theatre and room available on the same day, and insert a random compatible non-mandatory patient. Classes used: `Patient`, `Admission`, `Surgeon`, `SurgeonAvailability`, `OperatingTheatre`, `OperatingTheatreAvailability`, `Room`, `RoomAvailability`. Purpose: to insert a patient in a way that is more strongly guided by actual resource availability, especially surgeon and theatre constraints.
- Transformation 4: Remove a random non-mandatory patient. Classes used: `Patient`, `Admission`. Purpose: to free up resources while only sacrificing an optional patient, which helps repair overloaded or poor-quality schedules.
- Transformation 5: Remove a random patient. Classes used: `Patient`, `Admission`. Purpose: to make a larger change to the schedule and free capacity, even if that means removing a more important patient.
- Transformation 6: Remove a random non-mandatory patient, then insert a random non-mandatory patient. Classes used: `Patient`, `Admission`, `Room`, `OperatingTheatre`, `SurgeonAvailability`, `OperatingTheatreAvailability`, `RoomAvailability`. Purpose: to swap one optional patient for another, testing whether a different patient gives a better schedule without changing the total number of optional patients.
- Transformation 7: Remove a random patient, then insert a random non-mandatory patient. Classes used: `Patient`, `Admission`, `Room`, `OperatingTheatre`, `SurgeonAvailability`, `OperatingTheatreAvailability`, `RoomAvailability`. Purpose: to replace part of the current schedule with a new optional patient, allowing a more disruptive change in the set of admitted patients.

0.2 Patient Reassignment

- Transformation 8: Give a random admitted patient a random new compatible room. Classes used: `Patient`, `Admission`, `Room`, `RoomAvailability`. Purpose: to reassign a patient to a different feasible room in order to improve room usage or reduce incompatibilities.
- Transformation 9: Give a random admitted patient a random new compatible admission day. Classes used: `Patient`, `Admission`, `SurgeonAvailability`, `OperatingTheatreAvailability`, `RoomAvailability`. Purpose: to move a patient to a different feasible day in order to improve schedule balance or resource usage.
- Transformation 10: Give a random admitted patient a random new compatible theatre. Classes used: `Patient`, `Admission`, `OperatingTheatre`, `OperatingTheatreAvailability`. Purpose: to reassign a patient to a different feasible theatre in order to improve theatre allocation.
- Transformation 11: Change a patient's room, admission day, and theatre sequentially. Classes used: `Patient`, `Admission`, `Room`, `RoomAvailability`, `OperatingTheatre`, `OperatingTheatreAvailability`, `SurgeonAvailability`. Purpose: to make a broader coordinated change to one patient's allocation across several decision variables.
- Transformation 12: Change a patient's room and admission day sequentially. Classes used: `Patient`, `Admission`, `Room`, `RoomAvailability`, `SurgeonAvailability`, `OperatingTheatreAvailability`. Purpose: to jointly adjust accommodation and timing in a way that may improve feasibility or quality more than changing one variable alone.
- Transformation 13: Change a patient's admission day and theatre sequentially. Classes used: `Patient`, `Admission`, `OperatingTheatre`, `OperatingTheatreAvailability`, `SurgeonAvailability`, `RoomAvailability`. Purpose: to jointly modify the timing and surgical allocation of a patient while keeping the room unchanged.

0.3 Nurse Assignment

- Transformation 14: Add a random nurse to a random compatible room on a random compatible shift. Classes used: `Nurse`, `NurseWorkingShift`, `HospitalisationShift`, `Room`, `RoomShiftAssignment`. Purpose: to create a new nurse-to-room assignment where the nurse is available on that shift and satisfies the required skill and workload conditions.
- Transformation 15: Remove a random nurse from a random room on their schedule. Classes used: `Nurse`, `HospitalisationShift`, `RoomShiftAssignment`. Purpose: to remove an existing nurse-to-room assignment and allow the optimisation to repair or rebalance staffing decisions.
- Transformation 16: Remove a nurse from one room and add them to another. Classes used: `Nurse`, `NurseWorkingShift`, `HospitalisationShift`, `Room`,

RoomShiftAssignment. Purpose: to reallocate nursing capacity between room-shifts without changing the total

number of nurse assignments.

CONSTRAINTS

Patient Admission and Removal

- A patient must be assigned a feasible admission day.
- A patient must be assigned a feasible theatre, surgeon, and recovery room.
- The chosen surgeon must have capacity.
- The chosen operating theatre must have capacity.
- The chosen room must have capacity.
- For non-mandatory insertion moves, surgeon and nurse availability must exist.
- In transformation 3, the theatre and room must be available on the same day.
- In transformation 3, the inserted patient must be compatible with the chosen resources.
- Mandatory patients are prioritised in the initial feasible construction.

- Non-mandatory patients are not scheduled initially.
- Infeasible moves are permitted unless specified otherwise.
- Infeasible solutions receive a heavy penalty from the validator.

Nurse Assignment

- Nurses are assigned to rooms on specific shifts rather than directly to individual patients.
- A nurse can only be assigned to a room on a shift that belongs to their working schedule.
- Each room-shift has an associated workload, derived from the patients occupying that room during that period.
- A nurse assigned to a room-shift must have sufficient remaining workload capacity to cover that workload.
- A nurse assigned to a room-shift must also meet the required skill level for that room-shift.

OBJECTIVES

Patients Admission and Removal

- Maximise the number of scheduled patients.
- Minimise surgeon capacity violations.

- Minimise theatre capacity violations.
- Minimise room capacity violations.
- Minimise Standard deviation of room occupancy.
- Minimise the number of removed patients.